

STT 861: Theory of Probability & Statistics I

Lecture 12: Moments, Variance & Chebyshev

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Reference: Casella & Berger Ch. 2.3

Lecture 12 Outline

Today

- Moments: mean, variance, skewness, kurtosis
- Chebyshev's inequality
- Key formulas for Binomial, Exponential

Why Do We Need Moments?

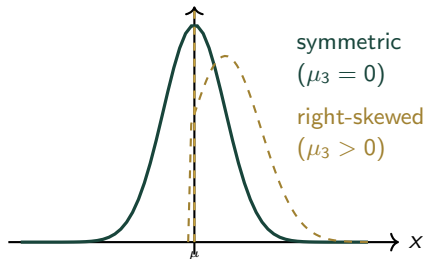
The Problem

Knowing the full pdf $f_X(x)$ is ideal, but:

- pdfs can be complicated
- We often only need a few key summaries
- Moments give us a **compact description** of a distribution's shape

What moments capture

- **1st moment:** center (μ)
- **2nd central moment:** spread (σ^2)
- **3rd central moment:** skewness
- **4th central moment:** tail heaviness



Key insight

The **mgf** encodes *all* moments in one compact function — and uniquely identifies the distribution.

Motivation — Moments in Practice

Insurance example

A claim amount X (in \$1000s).

- $EX = 5$: average claim is \$5,000
- $\text{Var}(X) = 25$: $\sigma = \$5,000$
- High 3rd moment: occasional very large claims (right skew)

The insurer needs **all three** to set premiums correctly.

Physics: moment of inertia

Variance = $E(X - \mu)^2$ is analogous to moment of inertia = $\sum m_i(x_i - \bar{x})^2$.

Larger variance $\leftrightarrow$ mass spread farther from center.

Why the MGF?

Computing EX^n by integration can be tedious.

The MGF shortcut: compute $M_X(t)$ once, then just differentiate:

$$EX = M'_X(0), \quad EX^2 = M''_X(0), \dots$$

Example (Exp(λ)): $M_X(t) = \frac{1}{1-\lambda t}$

- $M'_X(0) = \lambda \Rightarrow EX = \lambda$

- $M''_X(0) = 2\lambda^2 \Rightarrow \text{Var} = \lambda^2$

Much easier than integrating $\int_0^\infty x^2 \frac{1}{\lambda} e^{-x/\lambda} dx$!

Definition 2.3.1 — Moments

Definition 2.3.1 (CB)

The **n th moment** of X : $\mu'_n = EX^n$ The **n th central moment**: $\mu_n = E(X - \mu)^n$
where $\mu = \mu'_1 = EX$.

n	μ'_n	Meaning
1	$EX = \mu$	Mean
2	EX^2	2nd raw moment

n	μ_n	Meaning
1	$E(X - \mu) = 0$	always zero
2	$E(X - \mu)^2$	Variance
3	$E(X - \mu)^3$	Skewness
4	$E(X - \mu)^4$	Kurtosis

Relation: raw and central

$$\mu_2 = \mu'_2 - (\mu'_1)^2 = EX^2 - (EX)^2$$

$$\mu_3 = \mu'_3 - 3\mu'_2\mu + 2\mu^3$$

In practice, compute raw moments first (easier integrals), then convert.

Definition 2.3.2 & Shortcut Formula

Definition 2.3.2 (CB): Variance

$\text{Var}(X) = E(X - EX)^2$. Standard deviation: $\sigma_X = \sqrt{\text{Var}(X)}$.

Shortcut formula (Eq. 2.3.1):

$$\text{Var}(X) = EX^2 - (EX)^2$$

Proof:

$$\begin{aligned}\text{Var}(X) &= E(X^2 - 2X\mu + \mu^2) \\ &= EX^2 - 2\mu^2 + \mu^2 = EX^2 - \mu^2\end{aligned}$$

Interpretation

- Larger Var $\rightarrow$ more spread from mean
- $\text{Var}(X) = 0 \Rightarrow X$ is constant
- Units: $[\text{Var}] = [x]^2$; $[\sigma] = [x]$
- σ is easier to interpret

Jensen's connection

Since $g(x) = x^2$ is convex, Jensen gives $EX^2 \geq (EX)^2$, so $\text{Var}(X) \geq 0$. $\checkmark$

Example 2.3.3 — Exponential Variance

$X \sim \text{Exp}(\lambda)$: $EX = \lambda$

Compute EX^2 : (integration by parts twice)

$$EX^2 = \int_0^{\infty} x^2 \cdot \frac{1}{\lambda} e^{-x/\lambda} dx = 2\lambda^2$$

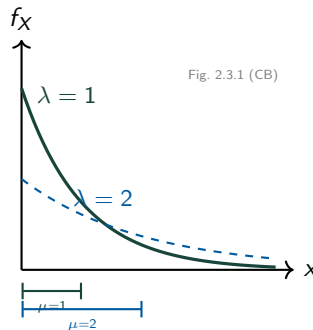
Shortcut:

$$\text{Var}(X) = EX^2 - (EX)^2 = 2\lambda^2 - \lambda^2 = \lambda^2$$

$$\sigma_X = \lambda = EX$$

Remarkable: $\sigma = \mu$ for Exponential

The coefficient of variation $\sigma/\mu = 1$ is a signature of the exponential — it characterises the memoryless property.



Theorem 2.3.4 — Variance of Linear Transformation

Theorem 2.3.4 (CB): $\text{Var}(aX + b) = a^2 \text{Var}(X)$

Proof: $\text{Var}(aX + b) = E[(aX + b) - (a\mu + b)]^2 = E[a(X - \mu)]^2 = a^2 \text{Var}(X). \quad \square$

Consequences

- Shifting: $\text{Var}(X + b) = \text{Var}(X)$
(adding a constant changes location, not spread)
- Scaling: $\text{Var}(aX) = a^2 \text{Var}(X)$
(doubling X quadruples the variance)
- Standardising: $Z = \frac{X - \mu}{\sigma}$
gives $EZ = 0$, $\text{Var}(Z) = 1$

Temperature example

If C = temperature in Celsius with $\text{Var}(C) = 4$ degrees²:

$$F = \frac{9}{5}C + 32:$$

$$\text{Var}(F) = \left(\frac{9}{5}\right)^2 \times 4 = \frac{81}{25} \times 4 \approx 12.96$$

The spread in Fahrenheit is $\left(\frac{9}{5}\right)^2$ times larger.

Example 2.3.5 — Binomial Variance (Part 1)

$X \sim \text{Binomial}(n, p)$: **Goal: find** $\text{Var}(X)$

We know $EX = np$. Need EX^2 . **Key identity:** $x^2 \binom{n}{x} = xn \binom{n-1}{x-1}$ (since $x \cdot \frac{n!}{x!(n-x)!} = n \cdot \frac{(n-1)!}{(x-1)!(n-x)!}$) So:

$$EX^2 = \sum_{x=1}^n x \cdot n \binom{n-1}{x-1} p^x (1-p)^{n-x}$$

Set $y = x - 1$: the sum becomes

$$= np \sum_{y=0}^{n-1} (y+1) \binom{n-1}{y} p^y (1-p)^{n-1-y} = np \left[\underbrace{(n-1)p}_{E[\text{Bin}(n-1, p)]} + \underbrace{1} \right] = np[(n-1)p+1] = n(n-1)p^2 + np$$

Example 2.3.5 — Binomial Variance (Part 2)

Completing the calculation

From Part 1: $EX^2 = n(n-1)p^2 + np$

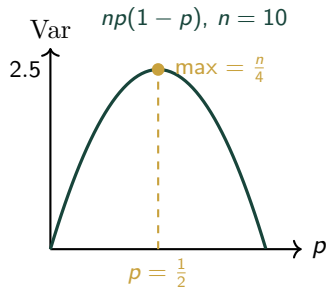
Apply shortcut $\text{Var}(X) = EX^2 - (EX)^2$:

$$\text{Var}(X) = n(n-1)p^2 + np - (np)^2 = -np^2 + np = np(1-p)$$

Summary: Binomial(n, p)

$$EX = np, \quad \text{Var}(X) = np(1 - p)$$

- Variance is maximised at $p = \frac{1}{2}$
- $\text{Var} = 0$ at $p = 0$ or $p = 1$ (certain outcomes)
- $n = 100, p = 0.5: \sigma = \sqrt{25} = 5$ successes



Definition 2.3.6 — Moment Generating Function

Definition 2.3.6 (CB): MGF

The **moment generating function** (mgf) of X is

$$M_X(t) = \mathbb{E} e^{tX}$$

provided $\mathbb{E} e^{tX}$ exists for all t in some open interval $(-h, h)$, $h > 0$.

$$M_X(t) = \begin{cases} \int_{-\infty}^{\infty} e^{tx} f_X(x) dx & X \text{ continuous} \\ \sum_x e^{tx} P(X = x) & X \text{ discrete} \end{cases}$$

Why e^{tX} ? The Taylor expansion reveals all moments

$$M_X(t) = \mathbb{E}e^{tX} = \mathbb{E} \sum_{n=0}^{\infty} \frac{(tX)^n}{n!} = \sum_{n=0}^{\infty} \frac{\mathbb{E}X^n}{n!} t^n = 1 + \mu'_1 t + \frac{\mu'_2}{2!} t^2 + \frac{\mu'_3}{3!} t^3 + \dots$$

The n th moment μ'_n is the coefficient of $t^n/n!$ in the Taylor expansion of $M_X(t)$.

Theorem 2.3.7 — MGF Generates Moments

Theorem 2.3.7 (CB): $EX^n = M_X^{(n)}(0)$

If X has mgf $M_X(t)$, then the n th moment equals the n th derivative of M_X at $t = 0$:

$$EX^n = \left. \frac{d^n}{dt^n} M_X(t) \right|_{t=0}$$

Proof: Differentiate $M_X(t) = \int e^{tx} f_X(x) dx$ under the integral:

$$\frac{d}{dt} M_X(t) = \int x e^{tx} f_X(x) dx = EX e^{tX} \xrightarrow{t=0} EX.$$

Repeating n times: $M_X^{(n)}(0) = EX^n$. $\square$

Recipe

(1) Find $M_X(t) = Ee^{tX}$. **(2)** Differentiate once: $M_X'(0) = EX$. **(3)** Differentiate twice: $M_X''(0) = EX^2$. **(4)** $\text{Var}(X) = M_X''(0) - [M_X'(0)]^2$.

Example 2.3.8 — Gamma MGF

$$X \sim \text{Gamma}(\alpha, \beta): f_X(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}$$

$$M_X(t) = \frac{1}{\Gamma(\alpha)\beta^\alpha} \int_0^\infty x^{\alpha-1} e^{-x(\frac{1}{\beta}-t)} dx$$

The integrand is the **kernel** of a $\text{Gamma}\left(\alpha, \frac{\beta}{1-\beta t}\right)$ pdf. Using $\int_0^\infty x^{\alpha-1} e^{-x/b} dx = \Gamma(\alpha)b^\alpha$:

$$M_X(t) = \frac{\Gamma(\alpha) \left(\frac{\beta}{1-\beta t}\right)^\alpha}{\Gamma(\alpha)\beta^\alpha} = \left(\frac{1}{1-\beta t}\right)^\alpha, \quad t < \frac{1}{\beta}$$

Mean from MGF:

$$M'_X(t) = \frac{\alpha\beta}{(1-\beta t)^{\alpha+1}} \Rightarrow EX = \alpha\beta$$

Special case

$\alpha = 1$: $\text{Exp}(\lambda)$ with $\beta = \lambda$

$$M_X(t) = (1 - \lambda t)^{-1} \checkmark$$

Example 2.3.9 — Binomial MGF

$X \sim \text{Binomial}(n, p)$

$$M_X(t) = \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1-p)^{n-x} = \sum_{x=0}^n \binom{n}{x} (pe^t)^x (1-p)^{n-x}$$

Apply the Binomial Theorem with $u = pe^t$, $v = 1 - p$:

$$M_X(t) = [pe^t + (1-p)]^n$$

Verify mean:

$$M'_X(t) = n[pe^t + (1 - p)]^{n-1} \cdot pe^t$$

$$M'_X(0) = np. \checkmark$$

Verify variance:

$$M''_X(0) = n(n-1)p^2 + np$$

$$\text{Var}(X) = np(1 - p). \checkmark$$

Poisson as limit

As $n \rightarrow \infty$, $p \rightarrow 0$, $np \rightarrow \lambda$:

$$M_X(t) = \left[1 + \frac{\lambda(e^t - 1)}{n}\right]^n \rightarrow e^{\lambda(e^t - 1)}$$

This is the $\text{Poisson}(\lambda)$ mgf — confirming the Poisson approximation to the Binomial.

Example 2.3.10 — Moments Don't Always Determine the Distribution

Two pdfs with identical moments

$$f_1(x) = \frac{1}{\sqrt{2\pi}x} e^{-(\log x)^2/2}, \quad x > 0$$

$$f_2(x) = f_1(x)[1 + \sin(2\pi \log x)], \quad x > 0$$

Claim: $EX_1^r = EX_2^r$ for all r .

Why: The extra term in EX_2^r becomes an odd-function integral $\rightarrow 0$ via the substitution $y = \log x - r$.

$f_1 \neq f_2$ but all moments agree!

Lesson

Knowing all moments $\neq$ knowing the distribution, *unless* the mgf also exists.

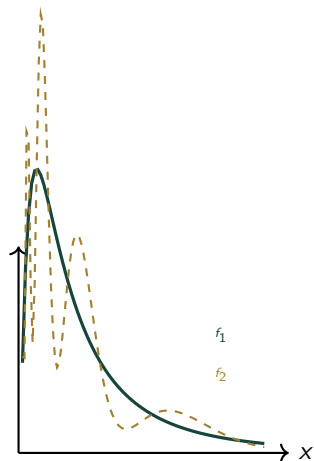


Fig. 2.3.2 (CB)

Theorem 2.3.11 — MGF Uniquely Determines the Distribution

Theorem 2.3.11 (CB)

Let F_X and F_Y be two cdfs all of whose moments exist.

- Ⓐ **Bounded support:** $F_X = F_Y \iff EX^r = EY^r$ for all $r = 0, 1, 2, \dots$
- Ⓑ **MGF uniqueness:** If $M_X(t) = M_Y(t)$ for all t in a neighborhood of 0, then $F_X = F_Y$.

Practical strategy: identify by MGF

Step 1: Compute $M_X(t) = Ee^{tX}$.

Step 2: Match to known mgf table.

Step 3: Conclude X has that distribution.

Example: $M_X(t) = (1 - 2t)^{-3}$
 $\Rightarrow X \sim \text{Gamma}(3, 2)$

Why moments may fail but MGF succeeds

The lognormal example (Ex 2.3.10) has all integer moments equal between f_1 and f_2 , but the lognormal mgf *does not exist* in any neighborhood of 0 (its tails are too heavy). So Thm 2.3.11(b) does not apply, and the moments don't determine the distribution.

Additional Example A — Exponential MGF

$$X \sim \text{Exp}(\lambda)$$

$$f_X(x) = \frac{1}{\lambda} e^{-x/\lambda}, x > 0.$$

$$M_X(t) = \frac{1}{\lambda} \int_0^{\infty} e^{tx} e^{-x/\lambda} dx = \frac{1}{\lambda} \cdot \frac{1}{\frac{1}{\lambda} - t} = \frac{1}{1 - \lambda t}$$

for $t < 1/\lambda$.

Moments:

$$M'_X(t) = \frac{\lambda}{(1 - \lambda t)^2} \Rightarrow EX = \lambda \checkmark$$

$$M''_X(t) = \frac{2\lambda^2}{(1 - \lambda t)^3} \Rightarrow EX^2 = 2\lambda^2$$

$$\text{Var}(X) = 2\lambda^2 - \lambda^2 = \lambda^2 \checkmark$$

MGF tells us the family

$$\begin{aligned} M_X(t) &= (1 - \lambda t)^{-1} \\ &= (1 - \lambda t)^{-\alpha} \text{ with } \alpha = 1 \\ &\Rightarrow \text{Exp}(\lambda) = \text{Gamma}(1, \lambda) \end{aligned}$$

Compare Gamma(2, λ)

$$\begin{aligned} M(t) &= (1 - \lambda t)^{-2} \\ EX &= 2\lambda, \text{Var} = 2\lambda^2 \end{aligned}$$

Additional Example B — Normal MGF

$X \sim N(\mu, \sigma^2)$: **Complete the square**

In the exponent of $e^{tx} f_X(x)$:

$$tx - \frac{(x - \mu)^2}{2\sigma^2} = -\frac{1}{2\sigma^2} [x - (\mu + \sigma^2 t)]^2 + \mu t + \frac{\sigma^2 t^2}{2}$$

So the integral becomes:

$$M_X(t) = e^{\mu t + \sigma^2 t^2 / 2} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sigma \sqrt{2\pi}} e^{-[x - (\mu + \sigma^2 t)]^2 / (2\sigma^2)} dx}_{=1} = \boxed{e^{\mu t + \sigma^2 t^2 / 2}}$$

Moments:

$$M'_X(0) = (\mu + \sigma^2 \cdot 0)e^0 = \mu = EX. \checkmark$$

$$M''_X(0) = \mu^2 + \sigma^2 = EX^2. \checkmark$$

$$\text{Var}(X) = \sigma^2. \checkmark$$

Normal MGF is the benchmark

If after computing $M_X(t)$ you get $e^{\mu t + \sigma^2 t^2 / 2}$, you can immediately conclude $X \sim N(\mu, \sigma^2)$ by Thm 2.3.11(b). This is the standard method for proving sums of normals are normal.

In-Class Activity — Section 2.3 (Q1 & Q2)

$X \sim \text{Uniform}(a, b)$, $f_X(x) = \frac{1}{b-a}$, $EX = \frac{a+b}{2}$.

(a) Compute $EX^2 = \int_a^b \frac{x^2}{b-a} dx$.

(b) Apply $\text{Var}(X) = EX^2 - (EX)^2$ to find $\text{Var}(X)$.

(c) Use Theorem 2.3.4 to find $\text{Var}(2X + 3)$ without further integration.

$P(X = x) = (1 - p)^{x-1}p$, $x = 1, 2, \dots$

(a) Show $M_X(t) = \frac{pe^t}{1 - (1 - p)e^t}$ for $t < -\log(1 - p)$.

(b) Differentiate and set $t = 0$: find $EX = M'_X(0)$. Confirm it equals $1/p$.

(c) Compute $M''_X(0)$ and find $\text{Var}(X)$.

In-Class Activity — Section 2.3 (Q3)

Use Theorem 2.3.11(b): if you recognise the mgf, you know the distribution.

(a) Suppose $M_X(t) = e^{3(e^t-1)}$.

Compare to the Poisson(λ) mgf: $M(t) = e^{\lambda(e^t-1)}$.

What distribution does X follow? Find EX and $\text{Var}(X)$ using M_X .

(b) Suppose $M_Y(t) = (1 - 2t)^{-3}$ for $t < \frac{1}{2}$.

Compare to the Gamma(α, β) mgf: $M(t) = (1 - \beta t)^{-\alpha}$.

What distribution does Y follow? Find EY and $\text{Var}(Y)$ by differentiating M_Y .

Solutions — Q1 & Q2

$$(a) EX^2 = \int_a^b \frac{x^2}{b-a} dx = \frac{a^2+ab+b^2}{3}$$

$$(b) \text{Var}(X) = \frac{a^2+ab+b^2}{3} - \frac{(a+b)^2}{4} = \frac{(b-a)^2}{12}$$

$$(c) \text{Var}(2X+3) = 4 \text{Var}(X) = \frac{(b-a)^2}{3}$$

$$(a) M_X(t) = \sum_{x=1}^{\infty} e^{tx} (1-p)^{x-1} p = pe^t \sum_{x=1}^{\infty} [(1-p)e^t]^{x-1} = \frac{pe^t}{1-(1-p)e^t} \text{ (geometric series, } |(1-p)e^t| < 1)$$

$$(b) M'_X(t) = \frac{pe^t}{[1-(1-p)e^t]^2}; \quad EX = M'_X(0) = \frac{p}{p^2} = \frac{1}{p}. \quad \checkmark$$

$$(c) M''_X(0) = \frac{2-p}{p^2}; \quad \text{Var}(X) = \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2}$$

Solutions — Q3

Compare to Poisson(λ) mgf $e^{\lambda(e^t-1)}$ with $\lambda = 3$.

By Thm 2.3.11(b): $X \sim \text{Poisson}(3)$.

$$M'_X(t) = 3e^t \cdot e^{3(e^t-1)}; \quad EX = M'_X(0) = 3.$$

$$M''_X(0) = 12; \quad \text{Var}(X) = 12 - 9 = 3.$$

For Poisson(λ): $EX = \text{Var}(X) = \lambda = 3$. ✓

Compare to Gamma(α, β) mgf $(1 - \beta t)^{-\alpha}$ with $\beta = 2, \alpha = 3$.

By Thm 2.3.11(b): $Y \sim \text{Gamma}(3, 2)$.

$$M'_Y(t) = \frac{6}{(1 - 2t)^4}; \quad EY = M'_Y(0) = 6 = \alpha\beta. \quad \checkmark$$

$$M''_Y(0) = 48; \quad \text{Var}(Y) = 48 - 36 = 12 = \alpha\beta^2. \quad \checkmark$$

Section 2.3 — Summary

Key Results

Variance shortcut	$EX^2 - (EX)^2$
Linear var	$\text{Var}(aX + b) = a^2 \text{Var}(X)$
$\text{Exp}(\lambda)$	$\mu = \lambda, \sigma^2 = \lambda^2$
$\text{Binom}(n, p)$	$\mu = np, \sigma^2 = np(1 - p)$
MGF	$M_X(t) = Ee^{tX}$
Moments	$EX^n = M_X^{(n)}(0)$
Uniqueness	same MGF $\Rightarrow$ same dist.

MGF Reference Table

$\text{Exp}(\lambda)$	$(1 - \lambda t)^{-1}$
$\text{Gamma}(\alpha, \beta)$	$(1 - \beta t)^{-\alpha}$
$N(\mu, \sigma^2)$	$e^{\mu t + \sigma^2 t^2 / 2}$
$\text{Binomial}(n, p)$	$[pe^t + (1 - p)]^n$
$\text{Poisson}(\lambda)$	$e^{\lambda(e^t - 1)}$